Deep Learning Based Regression Model using Neural Networks

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Abstract

The Challenge

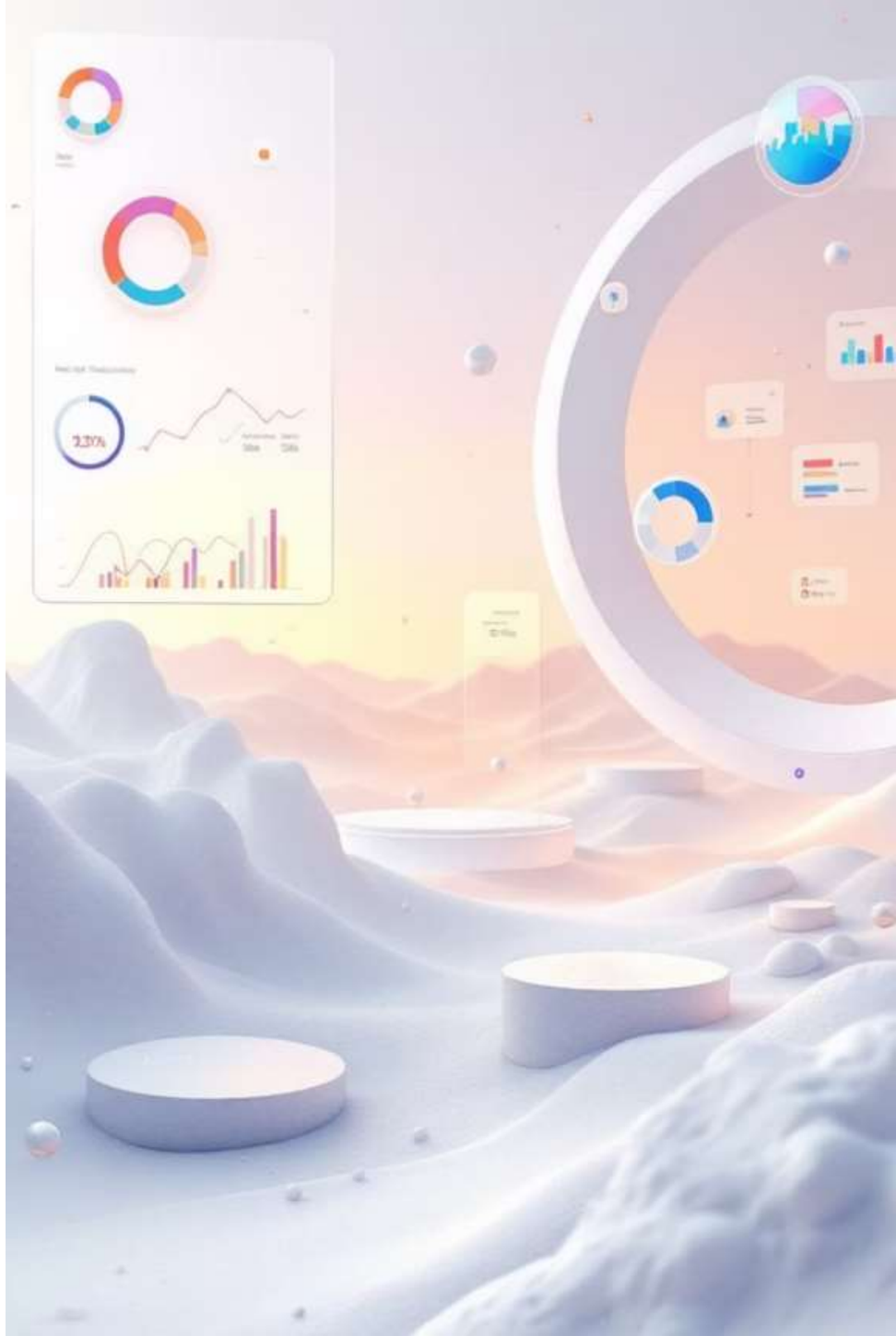
Traditional regression models struggle with complex and non-linear datasets used in real-world applications.

The Approach

Artificial Neural Networks (ANN) are used to learn complex patterns and improve prediction performance.

The Outcome

The model achieves improved prediction performance with reduced error (MAE and MSE).



Objectives

1

Data Preprocessing

Clean and normalize data

Prepare input features

2

Model Development

Design ANN model

Handle regression tasks

3

Training & Optimization

Train neural network

Reduce prediction error

4

Performance Evaluation

Evaluate using MAE and MSE

Measure model performance

INTRODUCTION

What Is Regression?

Regression is a supervised learning method used to predict continuous values.

Real-world applications:

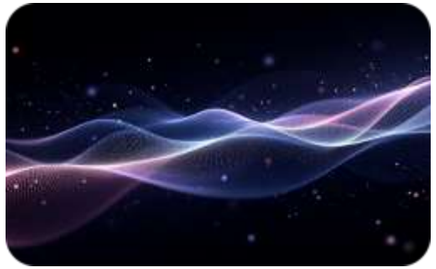
- House price prediction
- Stock market forecasting
- Medical outcome estimation

Traditional linear models fail when data relationships are **non-linear or high-dimensional**. Deep learning overcomes this by learning hierarchical feature representations automatically.

Literature Review

Linear Regression

Simple and interpretable, but limited to linear relationships. Performs poorly on complex, real-world datasets with non-linearity.



Support Vector Regression

Handles moderate non-linearity via kernel tricks, but requires careful tuning and struggles with very large datasets.

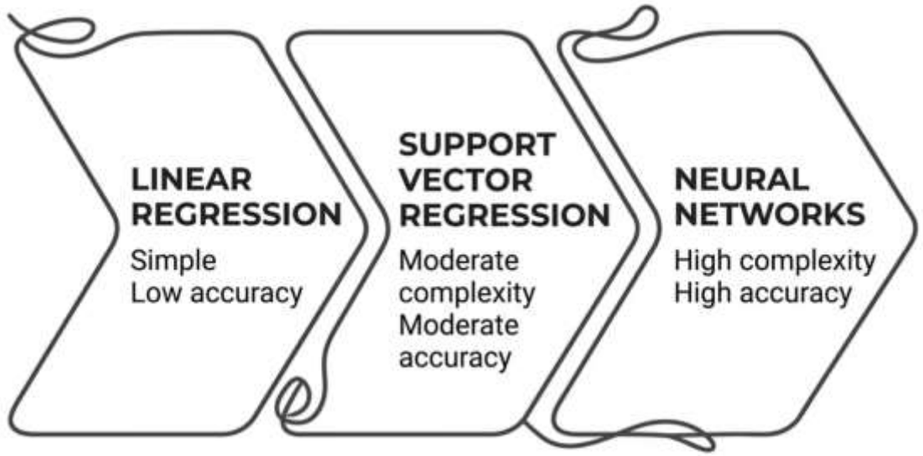
Neural Networks

Learns complex, non-linear patterns through multiple layers. Scales well with data volume and consistently outperforms classical methods.

EXISTING WORK

Limitations of Traditional Methods

Traditional regression methods struggle with complex and non-linear data in real-world applications.



Linearity Assumption

Assumes linear relationship between input and output

Feature Engineering Burden

Requires manual feature selection

Scalability Gaps

Performance decreases with large datasets

Proposed Work: ANN-Based Regression System

A fully connected Artificial Neural Network (ANN) is proposed to model non-linear regression problems through four core modules:

Data Preprocessing

Clean and normalize data
Handle missing values

Feature Preparation

Select important features
Split data for training and testing

Model Training

Train ANN model
Reduce prediction error

Prediction

Generate output values
Compare with actual data

System & Data Flow

Data flows through preprocessing and model stages to generate predictions.

Software & Tools

Python

Core programming language for model development and data pipelines.

TensorFlow

Open-source framework for building and deploying neural networks at scale.

Keras

High-level API for rapid prototyping of deep learning architectures.

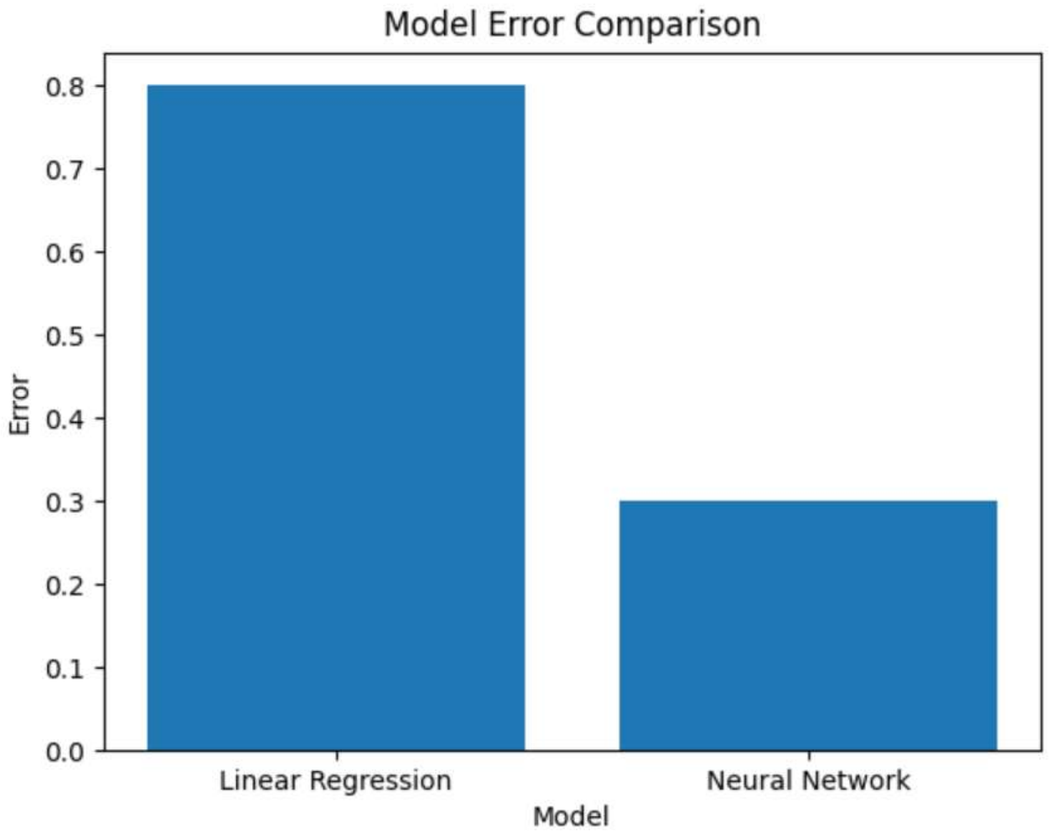
Pandas & NumPy

Data manipulation, numerical computation, and preprocessing utilities.

RESULTS

Performance Evaluation

The ANN model was evaluated using error metrics and compared with traditional regression methods.

- Neural Network shows lower error than Linear Regression
- Improved prediction performance observed
- Model captures non-linear relationships effectively

Error Comparison